Project Proposal & Specification Document

Project Information Overview

Attribute	Details
Project Title	StockSense: Explainable Stock Prediction and Research Report
Group Number	Group 13
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1. Executive Summary & Project Objectives

1.1 Executive Overview

Retail investing platforms like IBKR, moomoo, Plus500, Robinhood have made US stocks and ETFs easier to access, but beginner investors still face a difficult research task. They often have to reconcile information from multiple different avenues, such as price movement charts, market news, company analysis reports, volatility and personal risk tolerance. All this information, often, is spread across multiple disconnected sources. Raw charts and simplified ratings often do not provide transparency into their inner workings while board recommendations do not explain why a particular stock needs recommendation.

StockSense is a hybrid investment decision support system designed to guide beginner investors through a bounded, explainable research journey. By integrating machine learning predictive models with rule-based reasoning, the platform transforms fragmented and volatile market data into explainable investment insights. The web-based platform will not only provide personalised recommendations for stocks, but will also integrate a chat-bot architecture for users to generate further insights into a particular stock combining end-of-day market news as well as analyst and institutional research reports.

Users complete a short onboarding flow covering interests, risk profile, preferred sectors, and optional followed tickers. The product then presents a small, explainable research set, current market context, relevant news, a transparent 10-trading-day forecast with uncertainty, and a deterministic Explore, Monitor, or Caution signal. These labels guide research priority rather than trading action.

The system is built on a service-oriented FastAPI backend serving a responsive user interface. Its core architecture leverages neural networks (such as XGBoost and Transformers) for historical stock price forecasting, combined with a machine reasoning layer (Retrieval-Augmented-Generation and rule-based inference) that evaluates economic indicators and news sentiment to explain and validate recommendations.

Ultimately, StockSense resolves information overload and helps users make safer, more structured investment decisions. The initial scope limits the application to a curated selection of US stocks and ETFs and relies on publicly available market and news data. The proposal prioritises explainability, safety boundaries, and a measurable user experience over short-term prediction performance alone.

1.2 Core Project Goals

The goal of StockSense is to build an explainable stock discovery and decision-support system that helps beginner investors research US stocks and ETFs more confidently, without giving direct trading instructions.

The system aims to:

- Help beginner investors discover relevant stocks and ETFs based on their interests, risk profile, and preferred asset type.
- Combine market prices, company information, scheduled market news, and analyst research into one structured research view, using a RAG pipeline..
- Provide short-term stock movement forecasts with visible uncertainty, using predictive ML models, so users understand that predictions are probabilistic rather than guaranteed.
- Generate transparent research signals such as Explore, Monitor, and Caution, supported by clear explanations.
- Reduce information overload by summarising key market context, risk indicators, recent news, and relevant company information in one place.
- Support safer financial learning by avoiding direct Buy, Sell, or Buy More recommendations.

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2. Project Background

2.1 Problem Statement

- **Context:** The widespread adoption of low-cost trading platforms and fractional investing has dramatically lowered entry barriers, causing a surge in retail investor market participation.
- **The Problem:** Despite high data availability, retail investors lack the technical frameworks required to synthesize heterogeneous financial inputs, such as historical price series, technical indicators, qualitative news sentiment, and regulatory filings. Furthermore, traditional predictive tools output point-estimate forecasts without contextual uncertainty bounds, giving a false sense of precision while obscuring downside risk.
- **Impact:** This structural disconnect forces novice investors into cognitive fatigue, ungrounded speculation, or miscalibrated risk management. Ultimately, the lack of an auditable, explainable reasoning layer leads to poor decision-making and poor long-term portfolio performance.

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3. Industry Landscape & Market Analysis

3.1 Market Context & Overview

Most market platforms — Yahoo Finance, Apple Stocks, traditional brokerages like IBKR — expose abundant raw data and technical indicators, but their complex interfaces are unfriendly to beginners, and personalized advisory services are prohibitively expensive.

The broader AI-assisted investing category reflects this same split: it is estimated at roughly $14–19B in 2026, with multiple forecasts putting it past $50B by 2030, yet the field spans a $25,000/year institutional terminal at one end and free, general-purpose chatbots at the other, with few purpose-built tools designed for a guided beginner in between. (Chen and Sialm 2025, 25-30)

Most advisors now use AI in some part of their workflow, and adoption among younger investors is climbing, but generic chatbot answers still cannot show their reasoning or their uncertainty.

3.2 Market Research & Competitive Analysis

Player	Core approach (how it works)	Strength	Gap for a beginner	Source
BLOOMBERG TERMINAL **TIER 1 · INSTITUTIONAL**	AI-assisted natural-language search over decades of proprietary pricing, filings & news.	Unmatched data depth and real-time coverage.	$25,000+/yr built for professional analysts, not individual beginners.	bloomberg.com

ALPHASENSE **TIER 1 · INSTITUTIONAL**	NLP synthesis across earnings calls, broker research, expert transcripts & filings.	Fast, enterprise-grade qualitative research.	Enterprise pricing; no retail or beginner-facing product.	alpha-sense.com
SEEKING ALPHA(QUANT) **TIER 2 · AI RESEARCH**	Rules-based scoring across 5 factors: Value, Growth, Profitability, Momentum, EPS Revisions.	Transparent, auditable rating methodology.	Gives a rating, not a plain-language "why" or an uncertainty range.	seekingalpha.com
DANELFIN **TIER 2 · AI RESEARCH**	ML model on ~900 daily technical/fundamental/sentiment indicators → 1–10 AI Score.	Strong, published out-of-sample track record.	Score reflects a probability, not a full explanation of the reasoning.	danelfin.com
FINCHAT.IO **TIER 2 · AI RESEARCH**	NLP Q&A over filings, transcripts and financial statements.	Fast, natural-language lookup of company data.	Open-ended chat, not a bounded or structured research path.	finchat.io
TRENDSPIDER **TIER 2 · AI RESEARCH**	AI-automated trendline and pattern detection across timeframes.	Best-in-class technical signal detection.	Fundamentals and news sentiment are out of scope.	trendspider.com
BETTERMENT & WEALTHFRONT **TIER 3 · ROBO-ADVISORS**	Algorithmic ETF allocation (modern portfolio theory) plus automated tax-loss	Low fees, fully hands-off investing.	Advice is portfolio-level only, no individual stock insight.	betterment.com · wealthfront.com

	harvesting.			
ROBINHOOD **TIER 3 · BROKERAGE AI**	Generative AI "Stock Digests" summarizing news, technicals & analyst data.	Built into a brokerage app millions already use.	Explicitly labelled "not investment advice"; no fixed methodology.	robinhood.com
CHATGPT / GEMINI / PERPLEXITY **TIER 3 · GENERAL AI**	General LLM reasoning grounded by web search.	Free, flexible, conversational access.	Not finance-trained; answers aren't auditable or backed by a track record.	chatgpt.com · gemini.google.com perplexity.ai

3.3 The Market Gap: Why Current Financial Tools Fall Short

Limitations of current approaches include a total lack of explainability, separate and disconnected streams of qualitative and quantitative indicators, and the failure to provide a structured, bounded research path showing relevance, evidence, and uncertainty together. Black-box neural networks excel at short-term predictive accuracy but offer no deterministic audit trails, while conventional generative LLMs frequently hallucinate financial relationships when disconnected from structured domain ontologies.

Furthermore, current market tools analyze historical price trends and news sentiment in isolated siloes, preventing users from seeing how qualitative news directly drives numerical volatility. A large, fast-moving market therefore still lacks a dedicated, explainable research layer built for bounded beginner use—that gap is StockSense's opening.

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4. Project Scope & Technical Constraints

4.1 In-Scope Capabilities

- **User Onboarding & Profile Setup:** A simple questionnaire that records each user's risk level, preferred industries, choice of stocks or ETFs, and any companies they already follow.
- **Smart Stock & ETF Discovery:** Recommendations that introduce related companies and diversifying options based on user interests, while actively warning users against putting all their money into a single industry.
- **Short-Term Trend Forecasts:** Machine learning models that estimate general 10-day price ranges, complete with clear confidence levels and visible uncertainty bands rather than single, overconfident numbers.
- **Clear, Rule-Based Research Stances:** A transparent rule engine that blends price trends, market volatility, and recent news into three simple labels: **Explore** (worth looking into), **Monitor** (keep on your radar), or **Caution** (higher risk). Each label includes plain-English bullet points explaining why it was assigned.
- **Curated News & Economic Updates:** Relevant company news, industry headlines, and macroeconomic indicators (such as inflation and interest rates) saved and organized in the background to provide helpful context.
- **Interactive Web Dashboard & Helper Assistant:** A clean, easy-to-use web app featuring an onboarding flow, home overview, discovery screen, and company detail pages. It includes an optional assistant to answer simple questions about existing recommendations and company details.

4.2 Out-of-Scope Items

- **No Automated Trading:** The platform is strictly an educational research guide. It does not connect to brokerage accounts, manage user funds, or execute live trades.
- **No Live Day-Trading Data:** The system updates once per day after the US market closes, deliberately ignoring second-by-second price feeds and noisy social media chatter.
- **No Direct Buy/Sell Instructions:** The app avoids financial advice and never issues "Strong Buy", "Sell", or portfolio rebalancing orders.
- **No Whole-Market Scanning:** To keep data clean and manageable, the tool focuses entirely on a curated list of 30 to 50 well-known US stocks and ETFs across 5 to 8 major industries.

- **No Complex Portfolio Optimizers:** The project excludes black-box algorithms and multi-user tracking, keeping the guidance direct and easy to follow.

4.3 Technical & Operational Constraints.

- **End-of-Day Processing:** All calculations and data updates run in a single nightly batch after US markets close.
- **API Rate Caps:** Free data provider call limits require storing information locally rather than fetching it live on every page load.
- **Fixed Training Window:** Price models are trained on a five-year historical dataset using rolling past periods to test accuracy.
- **Predefined Asset List:** Calculations are locked to a preset list of assets to keep server storage light and dependable.
- **Pre-Tagged News Feeds:** The app sorts headlines using the data provider's existing topic tags rather than running its own text-analysis models.

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5. Data & Knowledge Pipeline Strategy

5.1 Data Sources & Knowledge Acquisition

StockSense will use a bounded set of market, company, news, analyst, and user-profile data sources to support personalised stock discovery, short-term forecasting, and explainable research signals. The project will focus on selected US stocks and ETFs rather than a global market, so that the prototype remains manageable, explainable, and suitable for beginner investors.

- **Primary Datasets:** Historical stock price data retrieved from the **Alpha Vantage API** and Yahoo finance's **yfinance ("Top Trending Stocks: US stocks with the highest interest today", n.d.)** library, including open, high, low, close, adjusted close, and volume data. These fields will be used to calculate returns, volatility, moving averages, momentum, and other forecasting features.

- **Market News Data:** Scheduled end-of-day and weekly market news will be collected using **Alpha Vantage API's ("Alpha Vantage Premium API Key", n.d.)** news and sentiment endpoint and, where suitable, NewsAPI's free news feed. Alpha Vantage already provides article-level and ticker-level sentiment fields, including article title, source, publication time, summary, topic relevance, overall sentiment score, overall sentiment label, ticker relevance score, ticker sentiment score, and ticker sentiment label.

- **Company and Asset Metadata:** Company name, ticker symbol, asset type, sector, industry, exchange, ETF category, and other descriptive attributes. This data will support filtering, content-based recommendation, and explanation generation.

- **Analyst and Institutional Research:** Third-party analyst reports, institutional research, company filings, P/E reports, will be used as contextual inputs for RAG. These sources will support qualitative understanding of company performance, sector outlook, valuation concerns, and business risks.

- **Macroeconomic Indicators:** Key economic indicators such as inflation, GDP growth, CPI, interest rates, and market-wide indicators may be used to provide broader market context where relevant.

- **User Profile Inputs:** Voluntary onboarding inputs such as risk profile, investment interests, preferred asset type, sector preferences, and optional followed tickers. StockSense will not collect brokerage credentials, transaction records, or sensitive financial account information.

- **Domain Expertise Integration:** Financial concepts such as volatility, valuation thresholds, diversification, sentiment, sector exposure, and risk level will be translated into expert-curated rules. These rules will support the system's explainable labels: Explore, Monitor, and Caution.

5.2 Data Processing & Knowledge Representation

1. **Ingestion & Preprocessing:** Historical price data, asset metadata, scheduled news updates, analyst research inputs, and macroeconomic indicators will be collected from selected sources. The data will be cleaned, normalised, and checked for missing values, duplicate records, inconsistent ticker symbols, and timestamp errors.
2. **Feature Engineering:** Raw market data will be transformed into structured features such as daily returns, rolling volatility, moving averages, momentum indicators, trading volume changes, and recent price trends. These features will support short-term forecasting and risk interpretation.
3. **News and Sentiment Preparation:** Alpha Vantage provides pre-calculated article-level and ticker-level sentiment scores. StockSense will ingest these scores, link them to relevant tickers, and use them as context features for explanation and research signals.
4. **Knowledge Extraction and Rule Design:** Compiling news sentiment scores, financial indicators, analyst context, sentiment labels, valuation metrics, diversification checks, and risk thresholds will be converted into rule-based logic. These rules will help determine when the system should assign Explore, Monitor, or Caution labels.

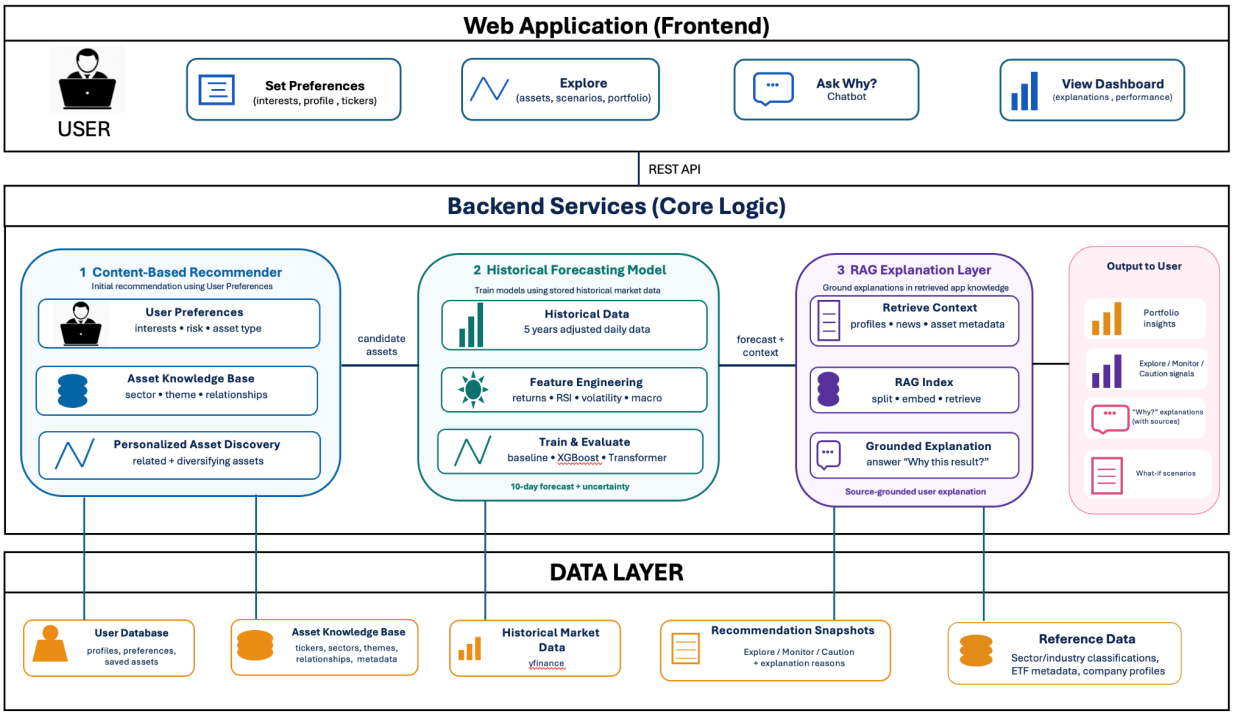
5. **Storage & Indexing:** Cleaned datasets and processed information will be stored in a structured schema using Mongo DB Atlas. The database will store news records, company analysis reports, user profile inputs, calculated features, and selected vector embeddings for retrieval. Relationships such as company-to-sector, stock-to-news, ETF-to-category, user-interest-to-asset, and ticker-to-sentiment will support personalised discovery, RAG-based retrieval, and explainable reasoning.

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6. Proposed System Architecture & Design

6.1 Architectural Overview

The proposed system is a personalised stock research and decision-support platform that integrates content-based asset discovery, historical market forecasting, and retrieval-augmented generation within a web application. The system is designed to help users discover relevant assets, view forecast information and receive source-grounded explanations rather than relying on unexplained model outputs.



1. User Interface / Web Application Layer

- Collects user interests, profile information and preferred tickers
- Allows users to set preferences
- Supports asset exploration
- Provides access to the “Ask Why?” chatbot
- Displays forecasts, explanations and performance information
- Presents personalised results through the dashboard

2. Backend

- Provides REST APIs between frontend and backend components
- Connects the:
    - Content-based recommender
    - Historical forecasting model
    - RAG explanation layer
    - Data layer
- Receives user requests from the frontend
- Returns processed results to the dashboard
- Coordinates the overall application workflow

(a) Content-Based Recommender

- Performs initial personalised asset discovery
- Uses user:
    - Interests
    - Risk preference
    - Preferred asset type
- Uses asset knowledge such as:
    - Sector
    - Theme
    - Asset relationships
- Matches user preferences with relevant assets
- Identifies related assets
- Identifies diversifying assets
- Produces candidate assets for further analysis

(b) Historical Forecasting Model

- Uses 5 years of adjusted daily historical market data
- Performs feature engineering using:
    - Returns
    - RSI
    - Historical volatility
    - Macroeconomic features
- Trains and evaluates:
    - Baseline model
    - XGBoost
    - Transformer
- Compares model performance
- Generates a 10-trading-day forecast
- Provides forecast uncertainty/confidence
- Supplies forecast information for candidate assets

(c) Rule-Based Reasoning Engine

- Acts as the final decision-support component
- Receives information from:
    - Content-Based Recommender
    - Forecasting Model
    - User profile
- Evaluates:
    - User risk tolerance
    - Recommendation context
    - Forecasted return
    - Forecast confidence
    - Historical volatility
    - Portfolio concentration
    - News events
    - Macroeconomic conditions
- Applies explicit deterministic rules
- Produces the final research signal:
    - EXPLORE
    - MONITOR
    - CAUTION
- Stores the reasoning/decision trace
- Makes the final signal transparent and explainable

(d) RAG Explanation Layer

- Provides source-grounded explanations
- Retrieves relevant context from:
    - User profiles
    - Financial news
    - Asset metadata
    - Company data
- Splits documents into suitable chunks
- Generates embeddings
- Stores/indexes embedded knowledge
- Retrieves relevant information for each query
- Generates explanations such as “Why this result?”
- Provides source-grounded responses to the user
- Acts as an explanation layer, rather than the forecasting model

(e) Output Layer

- Displays portfolio insights
- Displays :
    - Explore
    - Monitor
    - Caution
- Provides “Why?” explanations with sources
- Supports what-if scenarios
- Presents outputs through the personalised web dashboard

3. Data & Knowledge Layer

- User Database
    - Profiles
    - Preferences
    - Risk level
    - Holdings
    - Saved assets
- Asset Knowledge Base
    - Tickers
    - Sectors
    - Industries
    - Themes
    - Relationships
    - Company/ETF metadata
- Historical Market Database
    - 5-year adjusted daily data
    - Engineered features
- Recommendation Snapshots
    - Explore / Monitor / Caution
    - Forecast information
    - Rule-engine reasons

6.2 Reasoning Algorithm & Technique Justification

Technique	Evaluation Criteria	Selection Rationale
Content-Based Recommendation	Personalisation, interpretability, low dependency on other users	Used for initial asset discovery by matching the user's interests, risk and preferred asset type with asset sector, themes and relationships.
Baseline Forecasting Model	Simplicity, interpretability, benchmark performance	Establishes a reference performance level against which more advanced forecasting models can be evaluated.
XGBoost	Predictive performance, nonlinear modelling, suitability for engineered tabular features	Evaluated for predicting future stock returns using historical, technical and macroeconomic features.
Transformer	Temporal pattern modelling, ability to capture complex sequential dependencies	Evaluated as a deep-learning approach for learning patterns from historical market sequences.
RAG	Explainability, source grounding, contextual relevance	Retrieves relevant profile, news and asset information before generating explanations, reducing reliance on unsupported LLM knowledge.
Hybrid Approach	Personalisation + prediction + explainability	Combines content-based discovery, ML/deep-learning forecasting and RAG explanations so each technique performs a specialised role within the overall system.

7. Implementation Plan & Timeline

The system implementation maps directly to the modular architecture, breaking down into four self-contained development workstreams:

7.1 Core Workstreams

- **Workstream 1: Content-Based Recommender**
  - Builds automated pipelines pulling daily market data and asset metadata into a local cache.
  - Uses metadata embeddings and cosine similarity to match user risk and interests to candidate assets, applying correlation checks to prevent sector over-concentration.
- **Workstream 2: Feature Engineering & Baseline ML**
  - Constructs quantitative indicators from historical prices.
  - Trains statistical baselines and tuned XGBoost regressors to predict 10-day forward price trends, validated using rolling-window backtests to prevent lookahead bias.
- **Workstream 3: Deep Sequence Modeling & Risk Stances**
  - Implements deep sequential models to capture complex price dynamics.
  - Generates probabilistic uncertainty bands and maps predictions to clear research stances: **Explore**, **Monitor**, or **Caution**.
- **Workstream 4: RAG Explanation Layer & Web Application**
  - Constructs a vector store over curated news and company profiles to ground responses and prevent hallucination.
  - Deploys an interactive Streamlit dashboard backed by FastAPI, featuring forecast charts, stance summaries, and an "Ask Why?" conversational assistant.

7.2 Milestones & Execution Schedule

Phase	Key Deliverables	Estimated Completion Date(Tentative)
Phase 1: Setup & Ingestion	Curate asset list, build data cache, set up basic API shell.	2026-09-21
Phase 2: Core Models	Build candidate discovery, train baselines and XGBoost models, initialize vector store	2026-09-28
Phase 3: Deep ML & RAG	Train Transformer models, produce uncertainty bands, build RAG prompt chains.	2026-10-05
Phase 4: System Integration	Integrate workstreams, precompute recommendation snapshots, connect dashboard views	2026-10-12
Phase 5: Evaluation & Polish	Run benchmark metrics, finalise reports and repo.	2026-10-19

7.3 Proposed Tech Stack

Layer	Technology	Role in StockSense	Why This Choice
Presentation (Frontend Dashboard)	**React**	Builds the interactive dashboard: watchlists, signal cards, charts, and explanation panels the user browses.	A complete framework (routing, forms, dependency injection, HttpClient) that scales cleanly across a dashboard built and maintained by a team, with TypeScript built in from the start.
	TypeScript	Types every React component, service, and API model across the dashboard codebase.	Native to React's CLI and dependency-injection system; catches contract mismatches with the FastAPI backend at compile time.
	JWT Authentication	Protects every React route and API call: a login flow issues a signed token, route guards block unauthenticated pages, and an HTTP interceptor attaches the token as a bearer header on each request.	Stateless and scalable — FastAPI verifies the token on each call with no server-side session store, which fits a horizontally-scaled, containerized backend.
Application / API (Backend Services)	**Python 3.11+**	Core language for the FastAPI service layer and the entire	One language end-to-end from data science to API, no handoff or rewrite

		ML/reasoning pipeline it calls into.	between the ML team's models and the serving layer.
	FastAPI	Exposes REST/WebSocket endpoints that the React dashboard calls for signals, explanations, and portfolio data, and issues/validates the JWTs that protect them.	Async-first performance, automatic OpenAPI docs, and Pydantic validation, built for serving ML predictions with low latency.
Database (Storage & Knowledge)	**MongoDB**	Primary document store for user accounts, portfolios/watchlists, stock metadata, rule definitions, and the reasoning trace/audit log shown on the dashboard.	Schema-flexible documents fit StockSense's varied data shapes (user profiles, ML outputs, rule configs) better than a rigid relational schema, and map directly onto the JSON payloads FastAPI/Pydantic already produce.
	MongoDB Atlas Vector Search	Stores embeddings of financial knowledge, rules, indicator definitions, filings snippets. It runs the similarity search LangChain uses to retrieve supporting evidence for each recommendation (RAG).	Delivers RAG evidence-backed retrieval from inside the same MongoDB cluster, so the team operates one database technology instead of a separate graph store.
Reasoning & Intelligence (ML /	**XGBoost**	Gradient-boosted tree model for tabular forecasting — price-direction and risk	Fast to train, strong baseline on tabular financial data, and exposes feature

Explainability Engine)		scores from historical price, volume, and fundamental features.	importances that feed the explanation layer.
	Transformers/RNN/LSTM	Neural sequence models that learn non-linear temporal patterns across price/volume/news history to sharpen the directional signal.	State of the art for sequential pattern recognition, complements XGBoost's tabular strength with time-series depth.
	LangChain	Chains the ML forecast, a MongoDB Atlas Vector Search retrieval, and a plain-language write-up into one auditable reasoning trace shown to the user.	Purpose-built for composing multi-step model + retrieval pipelines. The mechanism behind StockSense's “explainable, not black-box” promise, now grounded in vector-retrieved evidence.
Deployment & Infrastructure	**Docker**	Containerizes the FastAPI service, the ML/reasoning pipeline, and the MongoDB client layer so every environment runs identically.	Reproducible builds from a developer laptop to production; each service scales or restarts independently.
	Azure(container hosting)	Hosts the containerized services and the React dashboard's static build behind a public URL.	

8. Expected Outcomes, Evaluation & Risk Management

8.1 Success Metrics & Evaluation Strategy

A. Forecasting Model Evaluation

- Evaluate the 10-trading-day return forecast
- Compare:
    - Baseline model
    - XGBoost
    - Transformer
- Metrics:
    - MAE – average magnitude of prediction error
    - RMSE – penalises larger prediction errors
    - Directional Accuracy – percentage of correctly predicted upward/ downward movements
- Compare all models against the baseline
- Select the best-performing model for final deployment

B. Content-Based Recommender Evaluation

- Evaluate the relevance and personalisation of recommended assets
- Evaluate recommendations based on:
- User interests
- Risk profile
- Preferred asset type
- Diversification
- Metrics:
    - Precision@K – percentage of top-K recommended assets that are relevant to the user
    - NDCG@K – measures whether the most relevant assets are ranked higher
    - Risk Compatibility Rate – percentage of recommendations aligned with the user's risk profile
    - Diversification Score – measures diversity across sectors/themes in the recommended assets
- Test using multiple predefined user profiles

- Verify that recommendations change appropriately with different user preferences

C. Rule-Based Engine Evaluation

- Evaluate the correctness of Explore / Monitor / Caution signals
- Test different combinations of:
    - User risk
    - Recommendation context
    - Forecast and confidence
    - Historical volatility
    - Portfolio concentration
    - News/macro context
- Metrics:
    - Rule Accuracy – percentage of predefined scenarios producing the expected signal
    - Rule Consistency – verifies that identical inputs always produce identical outputs
    - Decision Trace Completeness – percentage of signals with a complete explanation of the rules triggered
- Perform boundary testing around rule thresholds
- Verify that every generated signal can be traced back to explicit rules

D. RAG Explanation Evaluation

- Evaluate the quality and grounding of generated explanations
- Test questions such as:
    - “Why is this asset ‘Explore’?”
    - “Why did I receive ‘Caution’?”
    - “What information influenced this result?”
- Metrics:
    - Retrieval Precision@K – percentage of retrieved chunks relevant to the question
    - Faithfulness / Groundedness – degree to which generated claims are supported by retrieved evidence
    - Answer Relevance – measures whether the explanation directly answers the user's question
    - Citation Accuracy – percentage of citations that correctly support the generated claims
- Check for unsupported or hallucinated information

- Verify that RAG explains the existing signal rather than changing it

E. Overall System Evaluation

- Evaluate the performance and reliability of the complete application
- Test the complete workflow:
    - User preferences
    - Asset recommendation
    - Forecast generation
    - Signal generation
    - RAG explanation
    - Dashboard display
- Metrics:
    - Response Time – time required to return system results
    - API Success Rate – percentage of API requests completed successfully
    - Cache Hit Ratio – percentage of requests served using cached external data
- Perform end-to-end functional testing
- Verify successful integration between frontend, backend, models, databases and RAG

8.2 Risk Register & Mitigation Strategies

Risk 1: API Rate Limit Exhaustion

- **Risk:** Free APIs such as Alpha Vantage have limited daily requests (capped at 25 daily requests). (“Alpha Vantage Premium API Key”, n.d.)
- **Mitigation:** Cache historical prices, news and macroeconomic data locally; use scheduled data updates instead of repeated API calls; restrict the prototype to a selected stock/ETF universe.

Risk 2: Poor Forecasting Accuracy

- **Risk:** Stock prices are noisy and unpredictable, which may result in inaccurate 10-day forecasts.
- **Mitigation:** Compare the baseline, XGBoost and Transformer using walk-forward validation; include uncertainty estimates; deploy only the best-performing model.

Risk 3: Model Overfitting

- **Risk:** Forecasting models may perform well on training data but poorly on unseen market data.
- **Mitigation:** Use temporal train/validation/test splits, walk-forward validation, regularisation and comparison against a simple baseline.

Risk 4: Market Volatility / Regime Changes

- **Risk:** Sudden market conditions may reduce the reliability of historical patterns.
- **Mitigation:** Include volatility and macroeconomic indicators, provide forecast uncertainty and allow the rule engine to produce **Monitor/Caution** when uncertainty or risk is high.

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9. Conclusion & Expected Impact

9.1 Conclusion

This proposal outlines the design, methodology, and implementation roadmap for StockSense, an intelligent decision-support system tailored for retail investors. Rather than relying on isolated metrics or opaque black-box models, StockSense integrates three modular layers into a unified pipeline: profile-aligned asset screening, multi-model trend forecasting with uncertainty estimation, and grounded retrieval-augmented explanations. Developed as a modular web application delivered across a structured five-phase timeline, the project demonstrates how quantitative predictive techniques and transparent natural language reasoning can work together to provide accessible, auditable investment analysis.

9.2 Impact

- **Cuts Through Market Noise:** Replaces overwhelming data dumps and disjointed financial tools with a clear, guided pipeline that helps users quickly identify relevant assets.
- **Replaces Black Boxes with Context:** Moves past bare price predictions by pairing trend estimates with clear research stances and auditable "**why**" explanations backed by real market sources.
- **Promotes Responsible Investing:** Visually flags forecasting uncertainty, discouraging blind speculation and helping everyday investors evaluate risk before taking action.
- **A Functional Blueprint for Hybrid AI:** Demonstrates how recommendation, time-series modeling, and generative retrieval can be realistically unified into a responsive, accessible prototype.

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10. Supplementary Materials & Governance

10.1 AI Assistance Disclosure

- **Tools Used:** Gemini, ChatGPT
- **Scope of Use:** Proposal layout structuring, descriptive content generation

10.2 Key References & Documents

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